

The 2-adic deficit of the gcd permanent: two odd-permanent engines and a parity mechanism

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Abstract

Let $a(n) = \text{per}[\text{gcd}(i, j)]_{1 \leq i, j \leq n}$ and $D(n) = v_2(n!) - v_2(a(n))$ its 2-adic deficit below the factorial. Exact data at the peaks $n = 2^k + 1$ gives $v_2(a) = 3, 5, 11, 25$ ($k = 2, 3, 4, 5$), i.e. $D(2^k + 1) = 2k - 4$. We explain this value *mechanistically*. Two exact 2-adic valuation theorems for odd-integer permanents—one for size 2^a driven by the first permenal symmetric function π_1 , one for size $2^a + 1$ driven by π_2 —combine to give, for the two factors of the weight-0 grade $N_0 = \sum_v \text{per}(A_v) \text{per}(B_v)$, the sum $v_2(\text{per } A_v) + v_2(\text{per } B_v) = 2^k - 2k + 3$ at every v with $\pi_1(A_v), \pi_2(B_v)$ both odd. A leading-term/achiever-parity rule then gives $v_2(N_0) = 2^k - 2k + 3$ iff the number of such v is odd. That achiever count is computable with *no permanents* (an $O(n \log n)$ divisor sieve); it is odd for $k = 4, 5, 7, 8, 11, 14, 16, 17, 18$ within $k \leq 21$ (roughly half, recurring), giving $D_{N_0}(2^k + 1) = 2k - 4 \rightarrow \infty$. Thus the deficit of N_0 is (empirically, on a proven mechanism) **unbounded, of order $2 \log_2 n$** . The two engines and the reduction are rigorous; the sole open step—that the achiever parity is odd infinitely often—is a parity-of-a-divisor-correlation statement that behaves pseudorandomly and which we leave as a conjecture.

1 The two odd-permanent engines

For an $s \times s$ integer matrix M all of whose entries are odd, write $M = 2e + J$ with J the all-ones matrix and $e = (M - J)/2$. The column-multilinear expansion of the permanent (`OddPerm.permanent_two_mul_add_one`) is

$$\text{per}(2e + J) = \sum_{k=0}^s 2^k (s - k)! \pi_k(e), \quad \pi_k(e) = \sum_{|R|=|S|=k} \text{per}(e[R, S]), \quad (1)$$

the π_k being the permenal symmetric functions ($\pi_0 = 1$, $\pi_1 = \sum_{i,j} e_{ij}$). Writing $v_2(m!) = m - s_2(m)$ (s_2 = binary digit sum), term k has valuation $k + (s - k) - s_2(s - k) + v_2(\pi_k) = s - s_2(s - k) + v_2(\pi_k)$.

Theorem 1 (*A-engine, size 2^a*). *Let $s = 2^a$ with $a \geq 2$ and $\pi_1(e)$ odd. Then $v_2(\text{per}(2e + J)) = 2^a - a$ exactly.*

*Companion to *The permanent of the GCD matrix*. Data/scripts in `code/` of the **rama-notes** repository; the odd-permanent lower bound `OddPerm.two_pow_dvd_permanent_odd` and the $c = 1$ rate are machine-checked in `RamaLean/`. Correspondence: vicobonfioli@gmail.com.

Proof. In (1): the $k = 1$ term has $v_2 = 2^a - s_2(2^a - 1) + v_2(\pi_1) = 2^a - a$ (as π_1 odd); the $k = 0$ term has $v_2 = 2^a - 1 > 2^a - a$; and for $2 \leq k \leq 2^a$ one has $2^a - k \in [0, 2^a - 2]$, every element of which has at most $a - 1$ binary ones (only $2^a - 1$ has a), so $s_2(2^a - k) \leq a - 1$ and the term has $v_2 \geq 2^a - a + 1$. The $k = 1$ term is thus the unique minimum-valuation term; no cancellation at level $2^a - a$, hence $v_2(\text{per}) = 2^a - a$. $\square$

Theorem 2 (*B-engine, size $2^a + 1$*). *Let $s = 2^a + 1$ with $a \geq 3$ and $\pi_2(e)$ odd. Then $v_2(\text{per}(2e + J)) = 2^a + 1 - a$ exactly.*

Proof. Now $\max_{0 \leq k \leq s} s_2(s - k) = a$, attained *uniquely* at $k = 2$ (where $s - k = 2^a - 1$, the only integer $\leq 2^a + 1$ with a ones). So the $k = 2$ term has $v_2 = 2^a + 1 - a + v_2(\pi_2) = 2^a + 1 - a$ (π_2 odd), while every other k has $s_2(s - k) \leq a - 1$, hence $v_2 \geq 2^a + 2 - a$. Unique minimum, no cancellation, so $v_2(\text{per}) = 2^a + 1 - a$. $\square$

Theorem 1 is the tight matching upper bound to the machine-checked lower bound $v_2(\text{per}) \geq s - \lfloor \log_2 s \rfloor - 1$; both engines are exact, and were verified exhaustively on random odd matrices for $a \leq 4$.

2 The weight-0 grade and the mechanism

For odd n the weight-0 grade of $a(n) = \sum_{w \geq 0} 2^w N_w$ factors as $N_0 = \sum_v \text{per}(A_v) \text{per}(B_v)$, where, with $m = \lfloor n/2 \rfloor$, the sum runs over odd values v , A_v is the $m \times m$ all-odd matrix on even positions $\times$ (odd values $\setminus \{v\}$), and B_v the $(m+1) \times (m+1)$ all-odd matrix on odd positions $\times$ (even values $\cup \{v\}$). At the peaks $n = 2^k + 1$ we have $m = 2^{k-1}$, so A_v has size 2^{k-1} and B_v size $2^{k-1} + 1$: *exactly the two engine sizes*. When $\pi_1(A_v)$ and $\pi_2(B_v)$ are both odd, Theorems 1–2 give

$$v_2(\text{per } A_v) + v_2(\text{per } B_v) = (2^{k-1} - (k-1)) + (2^{k-1} + 1 - (k-1)) = 2^k - 2k + 3.$$

These are the minimum-valuation terms of N_0 (a π even only raises the valuation). Since all $\text{per}(A_v) \text{per}(B_v)$ are integers, the leading-term rule gives

$$N_0 \equiv 2^{2^k - 2k + 3} \cdot \#\{v : \pi_1(A_v), \pi_2(B_v) \text{ odd}\} \pmod{2^{2^k - 2k + 4}},$$

so $v_2(N_0) = 2^k - 2k + 3$ **iff the achiever count is odd**. This reproduces the measured $v_2(N_0) = 11, 25, 58$ at $k = 4, 5, 6$ exactly (the $k = 6$ value $58 > 55$ is the even-parity lift).

Proposition 3. *The achiever count $\#\{v : \pi_1(A_v), \pi_2(B_v) \text{ odd}\} \pmod{2}$ is computable in $O(n \log n)$ with no permanent evaluation, via $\pi_1(A_v) \equiv$ a divisor sum and an $O(s^2)$ closed form for $\pi_2(B_v) \pmod{2}$ reduced to Jordan-totient sums.*

Computed for $k = 4, \dots, 21$, the parity is odd at $k = 4, 5, 7, 8, 11, 14, 16, 17, 18$ (nine of eighteen), so $D_{N_0}(2^k + 1) = 2k - 4$ there, reaching 32 at $k = 18$ and setting new records at every odd-parity peak.

Corollary 4 (conditional). *If the achiever count is odd for infinitely many k , then $D_{N_0}(2^k + 1) = 2k - 4$ infinitely often, so the weight-0 deficit is unbounded, $D_{N_0} = \Theta(\log n)$ along the peaks.*

3 Status and the open problem

The engines (Theorems 1, 2) and the achiever-parity reduction are rigorous. What remains open is a single, sharply-posed arithmetic statement.

Conjecture 5. $\#\{v \text{ odd} \leq 2^k + 1 : \pi_1(A_v), \pi_2(B_v) \text{ both odd}\}$ is odd for infinitely many k .

Empirically this parity is $\approx \frac{1}{2}$ -dense and pseudorandom in k (no pattern in k , $s_2(k)$, or the binary expansion), which is why the conjecture, though it reduces to a computable divisor-sum correlation, resists a direct proof; the subfamily $k = 2^j$ is odd for $k = 4, 8, 16$ but its neighbours are odd too, so it is not evidently a tractable special case. For the *full* permanent $a(n)$ one needs additionally $v_2(a) = v_2(N_0)$ (weight-0 dominance), which holds at the peaks $k = 3, 4, 5$ with a grade-margin that grows.

Summary. The 2-adic deficit of the gcd permanent at $n = 2^k + 1$ equals $2k - 4$ through a transparent mechanism—two exact odd-permanent valuations plus a parity—rather than opaque cancellation. The deficit is unbounded conditional on a clean parity conjecture, and the mechanism is a permanent-free, $O(n \log n)$ predictor of $v_2(N_0)$.